

Estimation Activity 2

Team 10 | CS 4390/5388 | Campus Dining

Estimate dated September 28, 2026

Activity 2 · Step 1: Our selected slice

Selected slice and boundary

Question	Our slice
Slice name (one or two major parts)	Student visibility of campus dining options, locations, and sold-out items within each menu + mobile visibility.
Why we chose it	We want a one-stop solution for finding campus dining availability before students make a pointless commute.
Screens or forms where users enter data	Web browsers or mobile devices.
Reports, results, or lookups users see	Open dining options, available/sold-out items for each menu.
Data the slice stores	Per location times, menus and dining option availability.
Other systems it connects to	N/A
Out of the slice (not included)	Students can view a current crowd level and estimate wait time at each location.

Stories in the slice

Six user stories, totaling 22 points

User story	MoSCoW	Points
As a student, I want to view campus dining locations so that I can see what dining options are available	Must	3
As a student, I want to view each dining location's hours and open/closed status so that I know when I can visit.	Must	3
As a student, I want to view menu items for each dining location so that I can decide where to eat.	Must	3
As a student, I want to see whether menu items are available or sold out so that I do not waste time going somewhere for an unavailable item.	Must	5
As a student, I want to filter dining options based on dietary restrictions or preferences so that I can find suitable food.	Should	5
As a student, I want the dining information to work well on my phone so that I can quickly check dining options while on campus	Should	3
Total points		22

Activity 2 · Step 2: Our two methods

Our two methods: 1. Function Points 2. Use-Case Points

Why these two use different starting information: Function Points estimate the system based on what users enter, view, and what data the system stores, while Use-Case Points estimate the system based on the actors who interact with it and the use cases they perform.

Activity 2 · Step 3: The two estimates

Function points

Function-point counts and weights

Category	Count	Weight	Points
External Inputs (EI)	2	4	8
External Outputs (EO)	1	5	5
External Inquiries (EQ)	3	4	12
Internal Logical Files (ILF)	2	10	20
External Interface Files (EIF)	0	7	0
Total function points			45

Rate: low 6 and high 12 hours per FP

Assumption or source: Based on the suggested 6 to 12 hours per function point range provided in the activity.

Effort: $45 \text{ FP} \times 6 \text{ h} = 270 \text{ hours (low)}$; $45 \text{ FP} \times 12 \text{ h} = 540 \text{ hours (high)}$

Use-case points

Use-case-point calculation

Item	Calculation	Value
Actor weight (UAW)	Sum of actor weights	3
Use case weight (UUCW)	Sum of use case weights	25
Unadjusted (UUCP)	$UAW + UUCW$	28
Technical factor (TCF)	Assumed factor	1.0
Environment factor (ECF)	Assumed factor	1.0
Use-case points (UCP)	$UUCP \times TCF \times ECF$	28
Rate (hours per UCP)	Assumed productivity rate	20
Effort (hours)	$UCP \times \text{rate}$	560

UAW is the actor weight; UUCW is the use-case weight; UUCP is the unadjusted total. TCF and ECF are the technical and environment factors. The 1.0 factors and the rate of 20 hours per UCP are assumptions.

The worksheet gives category counts and actor/use-case weights, but it does not name the counted functions, actors, use cases, or transaction steps. These details still need to be recorded so the team can explain the totals.

Activity 2 · Step 4: Our team-size check

Team-size check recorded in the activity

Check	Yes / No
Our team is five or six people at most.	yes
We counted our communication paths: $n(n - 1) \div 2 = 6$	yes
We added explicit work for integrating the pieces of the slice.	yes
We did not assume that more people means proportionally more output.	yes
We estimated for the team we have.	yes
We adjusted, not copied, timelines from much larger teams.	no

With 4 people, communication paths = $4 \times (4 - 1) \div 2 = 6$.

The final check is marked "No" in the worksheet. Activity 2 does not use an analogous estimate or a larger team's timeline; it uses function points and use-case points. The integration-work check is marked "Yes," but no separate integration allowance is shown.

Activity 2 · Step 5: Why the estimates differ

Function points compared with use-case points

Comparison	Method 1	Method 2
Method	Function Points	Use-Case Points
Estimate in hours (low to high)	270 to 540 hours	560 hours
Middle value in hours	405 hours	560 hours
Input that drives it most	Productivity rate (hours per FP)	Rate (hours per UCP)
Is that input measured or assumed?	Assumed	Assumed

For Method 1, the middle value is: $(270 + 540) \div 2 = 405$ hours

For Method 2, 560 hours is both the estimate and the middle value.

Gap $560 \div 405 = 1.38$

Do your two methods depend on the same assumption?

No. Function Points depend on the system's inputs, outputs, inquiries, stored data, and productivity rate, while Use-Case Points depend on the actors, use-case complexity, and hours per UCP.

Which single input moves the answer most?

The Function Point productivity rate moves the answer the most. At 6 hours per FP, the estimate is 270 hours, while at 12 hours per FP, it increases to 540 hours.

What information would narrow the gap, and when could you get it?

Actual team productivity data would narrow the gap. Once development begins, we could measure the actual hours needed to complete features and use cases instead of relying on assumed productivity rates.

Our explanation of the gap

The Function Point method produced an estimate of 270 to 540 hours with a middle value of 405 hours, while the Use-Case Point method produced an estimate of 560 hours. The gap between the middle values is about 1.38. The estimates differ because Function Points measure inputs, outputs, inquiries, and stored data, while Use-Case Points measure actors and the complexity of their interactions with the system. Both methods also rely on assumed productivity rates because we do not yet have actual development data. Collecting real productivity data during development would help reduce this uncertainty.

Activity 2 · Step 6: Our dated estimate

Estimation appendix dated September 28, 2026

Field	Our estimate
Slice estimated	Student visibility of campus dining options, locations, menu availability/sold-out items, dietary filtering, and mobile visibility.
Range in hours (low to high)	270 to 560 hours
Expected hours: $(O + 4M + P) \div 6$, and why we chose M	438 hours. $O = 270$, $M = 450$, $P = 560$. $(270 + 4(450) + 560) \div 6 = 438.3$ hours. We chose 450 as the most likely value because it falls between the results of our two estimation methods.
Cost range and loaded rate used	\$16,200 to \$33,600 using an assumed loaded rate of \$60/hour.
Calendar estimate in weeks, and hours per week assumed	2.3 to 4.7 weeks. Assumes 4 team members $\times$ 30 productive hours/person/week = 120 team hours/week. Expected time $\approx$ 3.7 weeks.
Methods used	Function Points and Use-Case Points
Key assumptions	6 to 12 hours per FP; TCF = 1.0; ECF = 1.0; 20 hours per UCP; \$60/hour loaded rate; 30 productive hours/person/week.
Team size assumed	4 people
What would change this estimate	Changes in project scope, additional features or integrations, actual team productivity, use-case complexity, and changes to dining/menu data requirements.
Date of this estimate	September 28, 2026

Cost and calendar calculations

Cost: $270 \times \$60 = \$16,200$; $560 \times \$60 = \$33,600$. The \$60/hour loaded rate is an assumption.

Team capacity: 4 people $\times$ 30 productive hours/person/week = 120 team hours/week. Low: $270 \div 120 = 2.25$ weeks, rounded to 2.3. High: $560 \div 120 = 4.67$ weeks, rounded to 4.7. Expected: $438.3 \div 120 = 3.65$ weeks, rounded to 3.7.

This estimate covers the selected student-facing slice. Crowd levels and estimated wait times are outside it. The starter's 2-to-3-month first-release estimate covers a different boundary. Neither figure is a production schedule commitment.